

Exam

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Electricity and Electronics EBN111

11 June 2025

Assessment Information

Maximum marks:	93	Full marks:	90
Duration of test:	180 min + (5 min before and 15 min after for OCR answer sheet completion)	Open/Closed book:	Closed
Total number of pages (this page included)			32

Lecturers: Prof X. Ye, Mr S. Twala, Mr A. Rohde**Assistant Lecturers:** L. Hurter, J. Nepembe, I. Schilz, C. van Zyl**Important**

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR answer sheets provided. The question number **01** in this question paper refers to row 01 of the OCR answer sheet 00/01.
4. Read the instructions on the OCR answer sheet 00/01 carefully.
5. Where applicable, answers must include units.
6. The final answers must be written as real numbers and not as fractions.
7. Use at least 4 significant figures to write numbers, including complex numbers.
8. A complex number may be written in either rectangular or polar format.
9. Exact numbers in which decimal expansions are zero can be written without the zero expansions, for example 0 V instead of 0.000 V or 3.5 A instead of 3.500 A.

INSTRUCTIONS

STEP 1 (-5 min to 0 min): Complete the front page of OCR answer sheet 01, and also fill in your student number on additional OCR answer sheets.

STEP 2 (0 min to 180 min): Do your calculations on the question paper.

STEP 3 (180 min to 195 min): Carefully copy your answers to the OCR answer sheets. Be sure to include units where applicable.

- The Assessment ID is: **2025EBN111E01**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should have only one character, and the characters should not touch or cross the cell.
- **Do not use commas!** Please use full stops as decimal points.

Examples for complex numbers on OCR forms

5	∠	—	9	0	°				V	
6	.	9	∠	1	5	.	9	°	μ	A
1	2	—	j	8						
j	1	5							Ω	
—	1	.	5	+	j	8	.	2	m	V

Rectangular:The j is at the start of the imaginary number.The j is in the CENTRE of the cell.The j does NOT cross or touch cell boundaries!**Polar:**

The degree sign ° is directly after the angle value.

Section 1

Question 1 [2]

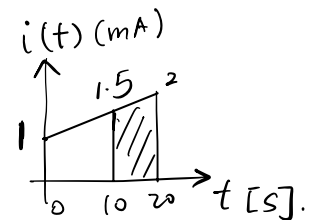
A series circuit has a voltage source and a $10\ \Omega$ resistor. The time-varying voltage across the resistor increases linearly from 10 mV to 20 mV between $t = 0$ s and $t = 20$ s.
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01	Calculate the total charge flowing between $t = 10$ s and $t = 20$ s. [2] <u>17.5 mC</u>
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$$\Delta V = \frac{20 - 10}{20} = 0.5\text{ mV/s}$$

$$\Delta i = \frac{\Delta V}{R} = \frac{0.5}{10} = 0.05\text{ mA/s}$$

$$i(t) = 0.05t + 1\text{ (mA)}$$



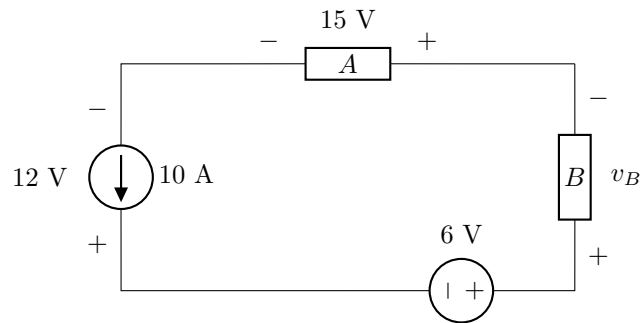
$$Q = \int_{10}^{20} i(t) dt = \int_{10}^{20} (0.05t + 1) dt$$

$$= \left(\frac{1}{2} \times 0.05 t^2 + t \right) \Big|_{10}^{20}$$

$$= \frac{1}{2} \times 0.05 (20^2 - 10^2) + (20 - 10)$$

$$= 17.5\text{ mC}$$

Questions 2 to 3 [3]

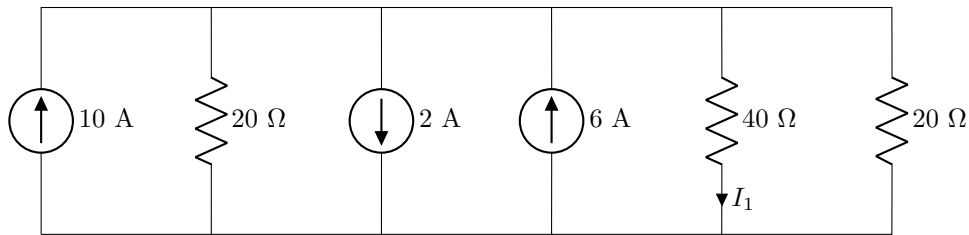


02	Determine the voltage v_B . [2]	3V
03	Determine the power delivered by the independent voltage source. [1]	-60W

02 $12 - 15 - v_B + 6 = 0 \Rightarrow v_B = 3V$

03 $P = 10 \cdot (-6) = -60W$

Question 4 [2]

04 Determine I_1 [2]

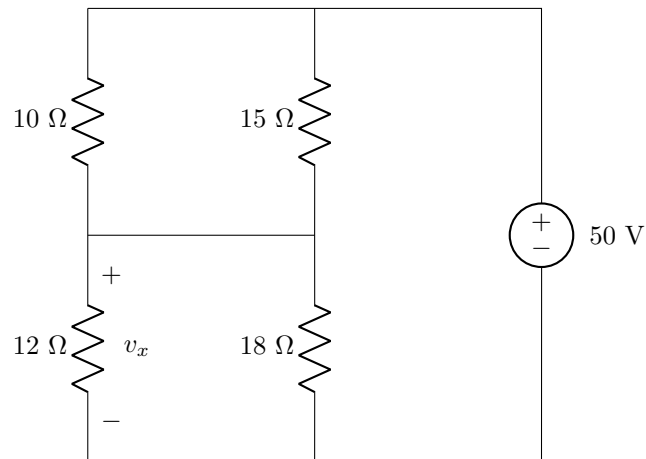
2.8 A

Handwritten solution:

$$(10 + 6 - 2)\text{ A} = 14\text{ A}$$
$$I_1 = 14 \cdot \frac{10}{50} = 2.8\text{ A}$$

The handwritten equivalent circuit diagram shows a single vertical loop. On the left is a 10 A current source pointing upwards. This is in series with a parallel combination of two resistors: a $40\ \Omega$ resistor and a $10\ \Omega$ resistor. The current through the $40\ \Omega$ resistor is labeled I_1 with a downward arrow.

Question 5 [2]

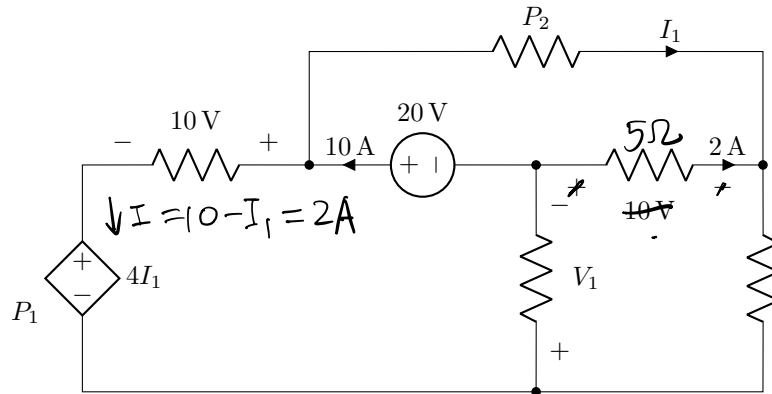
05 Determine v_x [2] $27.3V$

$$10 \parallel 15 = \frac{150}{25} = 6 \Omega.$$

$$12 \parallel 18 = \frac{12 \times 18}{30} = 7.2 \Omega.$$

$$v_x = 50 \cdot \frac{7.2}{6 + 7.2} = 27.3V.$$

Questions 6 to 7 [4]

Find P_1 and P_2 when $I_1 = 8$ A.

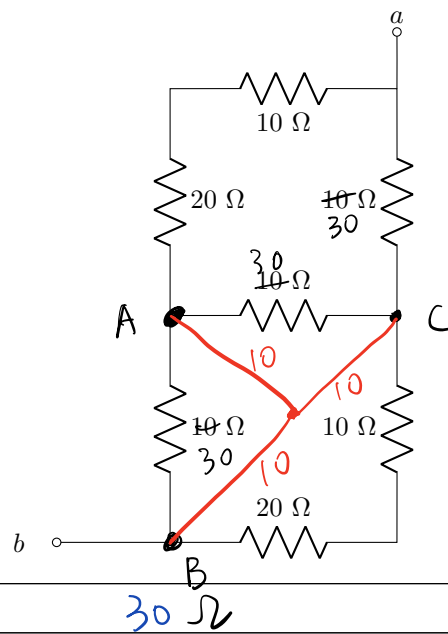
06	P_1 [2]	64W
07	P_2 [2]	240W

$$P_1 = 4I_1 \times 2A = 4 \times 8 \times 2 = 64W$$

$$V_{P_2} = 20 + 10 = 30V$$

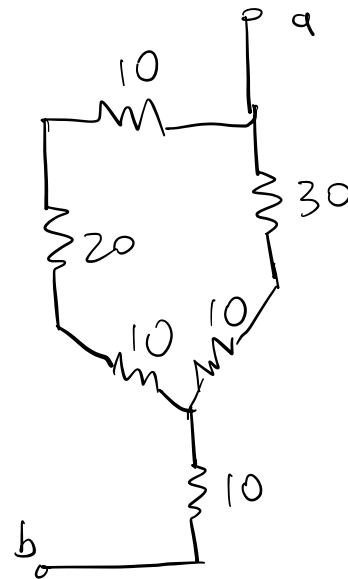
$$\Rightarrow P_2 = 30 \cdot I_1 = 240W$$

Question 8 [2]

08 Determine R_{ab} [2]

$\Delta ABC \rightarrow Y_{ABC}$, New circuit

$$\begin{aligned}
 R_{ab} &= 10 + 40 \parallel 40 \\
 &= 10 + 20 \\
 &= 30 \Omega
 \end{aligned}$$



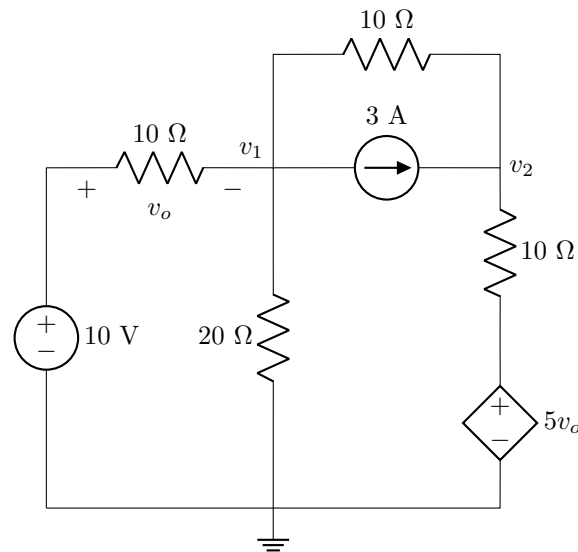
Questions 9 to 12 [4]

Use nodal analysis to set up two equations in the form

$$av_1 - 2v_2 = b,$$

$$cv_1 + dv_2 = 40.$$

Calculate the unknown parameters a , b , c , and d .



$$V_o = 10 - V_1$$

09	a [1]	5
10	b [1]	-40
11	c [1]	2
12	d [1]	1

$$V_1: \quad \frac{V_1 - 10}{10} + \frac{V_1}{20} + 3 + \frac{V_1 - V_2}{10} = 0 \quad \times 20$$

$$2V_1 - 20 + V_1 + 60 + 2V_1 - 2V_2 = 0$$

$$5V_1 - 2V_2 = -40$$

$$V_2: \quad -3 + \frac{V_2 - V_1}{10} + \frac{V_2 - 5V_o}{10} = 0$$

$$-30 + V_2 - V_1 + V_2 - 5(10 - V_1) = 0$$

$$4V_1 + 2V_2 = 80 \Rightarrow 2V_1 + V_2 = 40$$

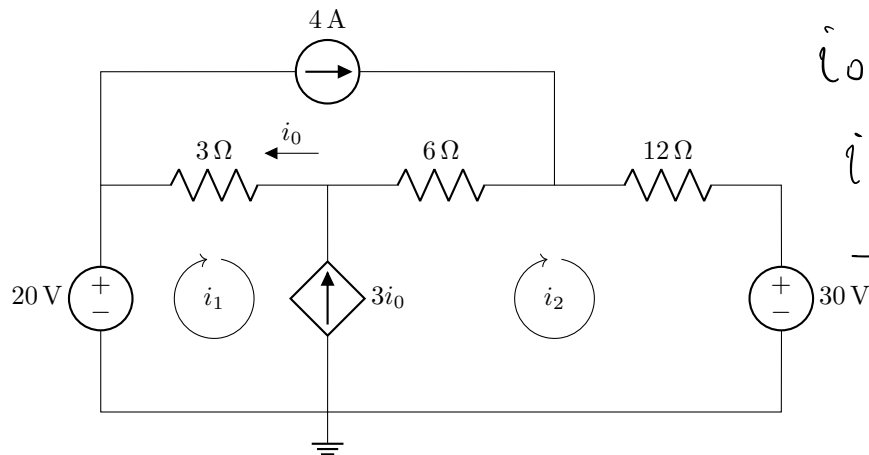
Questions 13 to 16 [4]

Use mesh analysis to set up two equations in the form:

$$3i_1 + ei_2 = f,$$

$$gi_1 + hi_2 = 12$$

Calculate the unknown parameters e , f , g and h .



$$i_0 = 4A$$

$$i_2 - i_1 = 3i_0 = 12$$

$$-i_1 + i_2 = 12$$

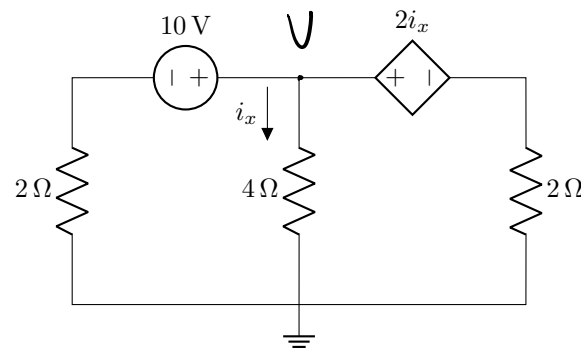
13	e [1]	18
14	f [1]	2
15	g [1]	-1
16	h [1]	1

super mesh:

$$-20 + 3(i_1 - 4) + 6(i_2 - 4) + 12i_2 + 30 = 0$$

$$3i_1 + 18i_2 = 20 + 12 + 24 - 30$$

$$= 26$$

Question 17 [2]Choose an appropriate approach between nodal and mesh analysis to determine i_x .

$$\hat{i}_x = \frac{V}{4}$$

17 i_x [2] 1.25 A

$$\frac{V-10}{2} + \frac{V}{4} + \frac{V-2 \cdot \frac{V}{4}}{2} = 0$$

$$\times 4$$

$$2V - 20 + V + 2V - V = 0$$

$$4V = 20$$

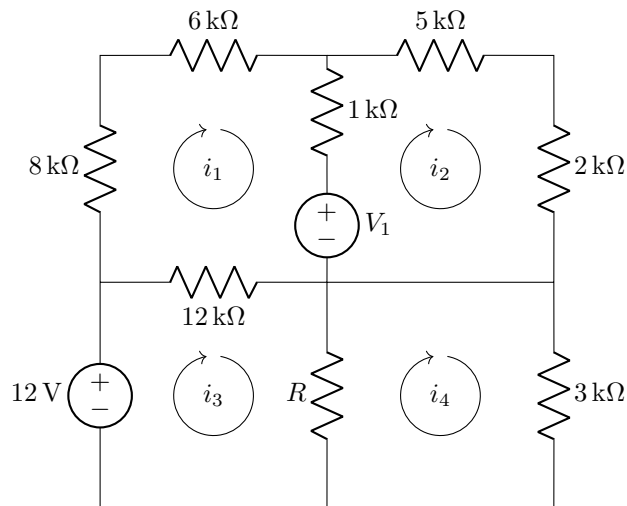
$$\Rightarrow V = 5V, \quad \hat{i}_x = \frac{5}{4} = 1.25A$$

Questions 18 to 19 [2]

The following matrix is constructed by inspection in terms of mesh analysis in the circuit.

$$\begin{bmatrix} 27000 & -1000 & -12000 & 0 \\ -1000 & 8000 & 0 & 0 \\ -12000 & 0 & 17000 & -5000 \\ 0 & 0 & -5000 & 8000 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 8 \\ -8 \\ 12 \\ 0 \end{bmatrix}$$

Determine R and V_1 in the circuit.

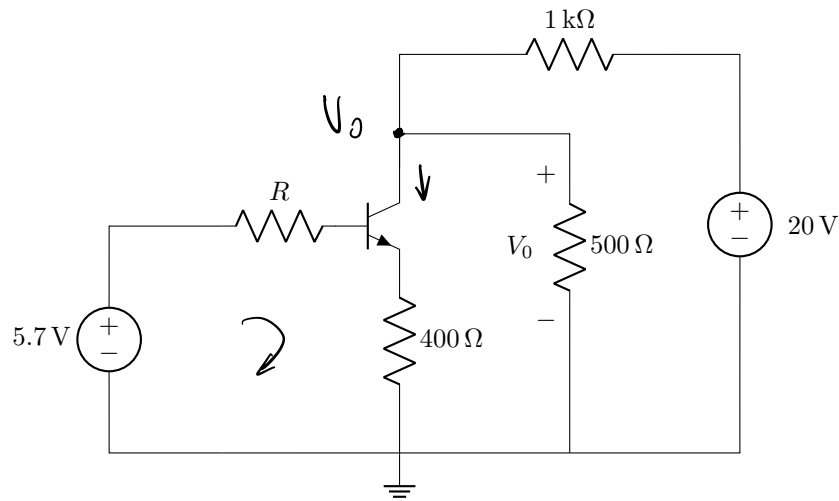


18 R [1] $5\text{ k}\Omega$

19 V_1 [1] -8 V

Questions 20 to 21 [4]

Given $I_B = 100 \mu\text{A}$, $\beta = 99$, and $V_{BE} = 0.7 \text{ V}$, determine R and V_0 .

20 R [2] $10 \text{ k}\Omega$ 21 V_0 [2] 10.1 V

(20) KVL: $-5.7 + I_B R + 0.7 + (I_B + \beta I_B) \cdot 400 = 0$

$$I_B R + 100 \times 400 I_B = 5$$

$$R = \frac{5 - 100 \times 400 I_B}{I_B} = \frac{5}{I_B} - 40 \text{ k}$$

$$= \frac{5 \times 10^6}{100} = 10 \text{ k}\Omega.$$

(21) KCL: $\beta I_B + \frac{V_0}{500} + \frac{V_0 - 20}{1000} = 0$

$$1000 \beta I_B + 2V_0 - V_0 - 20 = 0 \quad \times 1000$$

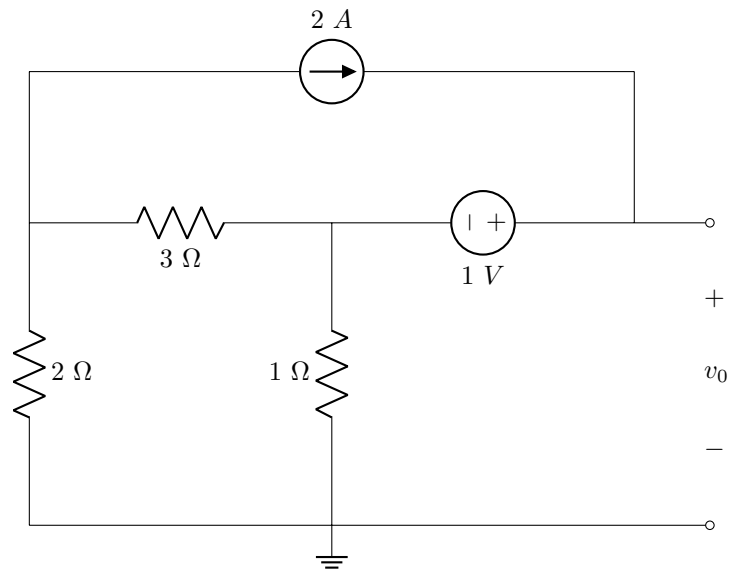
$$\Rightarrow V_0 = 20 - 1000 \times 99 \times 100 \times 10^{-6}$$

$$= 10.1 \text{ V}$$

Total Section 1: [31]

Section 2

Questions 22 to 23 [4]



22	Determine $v_0^{V_s}$, the independent contribution of voltage source to v_0 . [2]	1 V
23	Determine $v_0^{I_s}$, the independent contribution of current source to v_0 . [2]	1 V

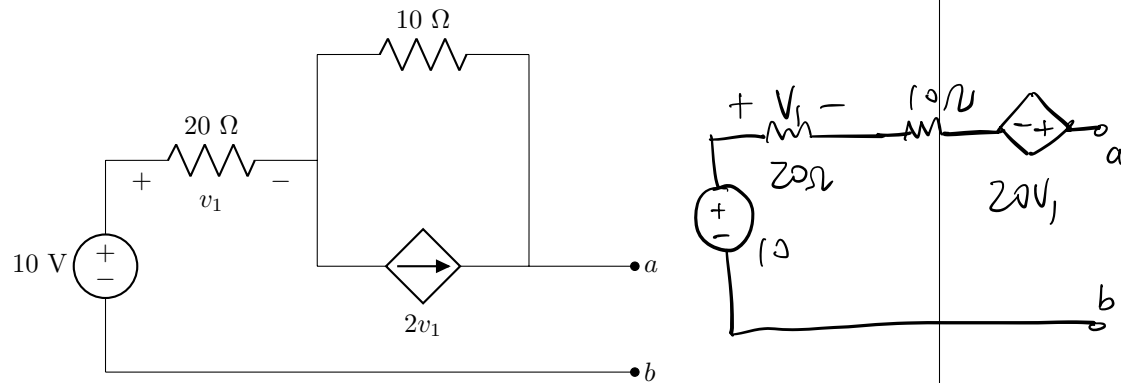
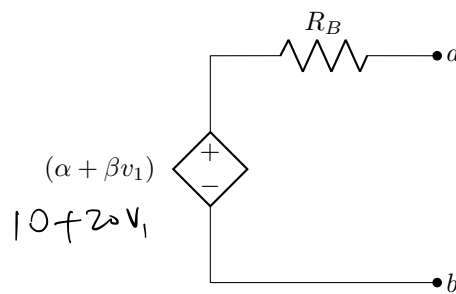
22) Switch off 2A. $V_o = 1V$

23) Switch off 1V, then use source transformation.

$$V_o = 6 \cdot \frac{1}{2+3+1} = 1V.$$

Questions 24 to 25 [2]

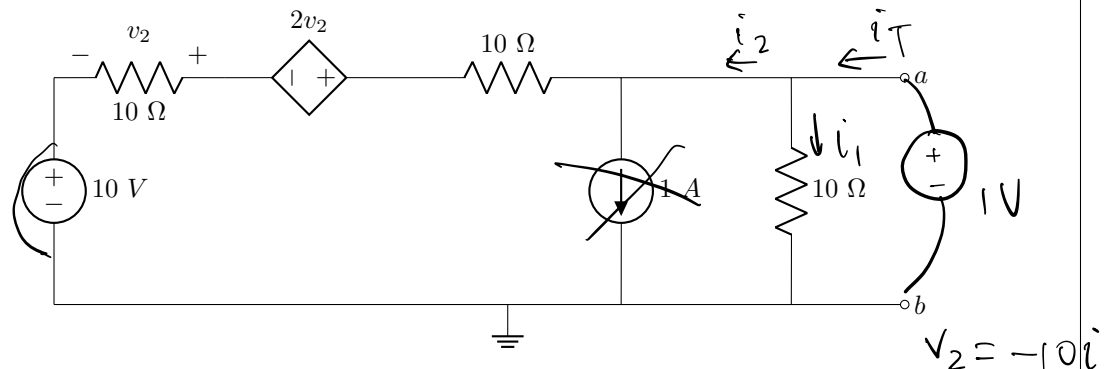
Use source transformation to obtain Circuit B so that it is equivalent to Circuit A with respect to terminals $a - b$.

Circuit A:**Circuit B:**

- | | | |
|-----------|--|---------------|
| 24 | Calculate α (unit required) [1] | 10 V |
| 25 | Calculate β (no unit required) [1] | 20 |

Questions 26 to 27 [4]

Determine the Thevenin equivalent voltage V_{TH} and the Thevenin equivalent resistance R_{TH} with respect to terminals a-b..

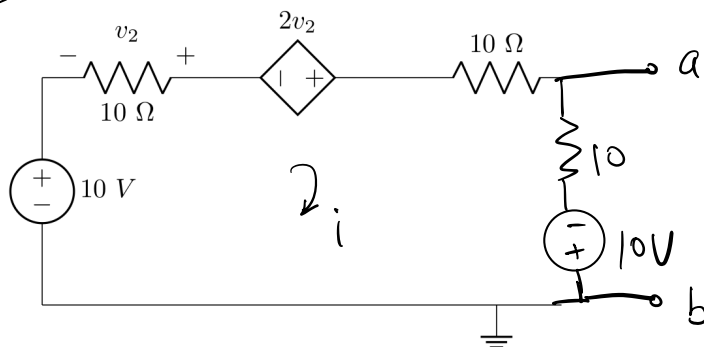


26 V_{TH} [2] $-6V$

27 R_{TH} [2] 8Ω

26 Use Source transformation

$$-10 + 10i - 2v_2 + 10i + 10i - 10 = 0$$



$$30i - 2(-10i) = 20$$

$$\Rightarrow 50i = 20 \quad i = 0.4A$$

$$V_{Th} = 10i - 10 = -6V.$$

27 Add 1V test voltage, switch off 10V & 1A.

$$i_1 = \frac{1}{10} = 0.1A$$

$$i_2 = \frac{1 - 2v_2}{20} = \frac{v_2}{10} \quad \times 20$$

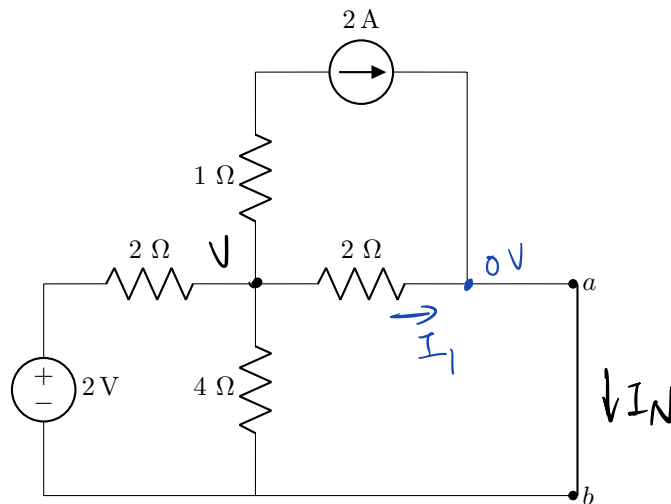
$$1 - 2v_2 = 2v_2 \Rightarrow v_2 = \frac{1}{4}$$

$$i_2 = \frac{1}{40} = 0.025, \quad i_T = i_1 + i_2 = 0.125$$

$$R_{Th} = \frac{1}{i_T} = 8\Omega$$

Questions 28 to 29 [4]

Find the Norton equivalent current I_N , and Norton equivalent resistance R_N , with reference to terminals $a - b$.

28 I_N [2]

1.6 A

29 R_N [2]

3.33 Ω

$$\textcircled{29} R_N = 2 + 2 \parallel 4 = 2 + \frac{8}{6} = 3.33 \Omega.$$

$\textcircled{28}$ use nodal analysis:

$$\frac{V-2}{2} + \frac{V}{4} + \frac{V}{2} + 2 = 0$$

$$2V - 4 + V + 2V + 8 = 0$$

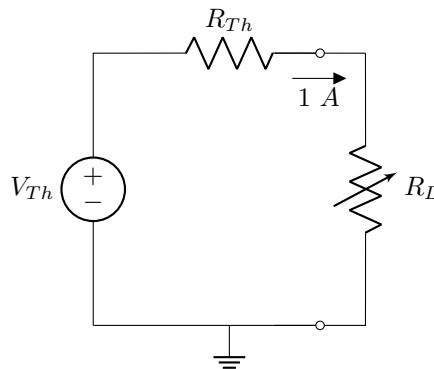
$$5V = -4 \Rightarrow V = -0.8$$

$$I_1 = \frac{-0.8}{2} = -0.4 \text{ A}$$

$$I_N = 2 + I_1 = 1.6 \text{ A}$$

Questions 30 to 31 [3]

The maximum power P_{max} delivered to R_L is given as 4 W. Determine the Thevenin equivalent voltage V_{Th} .



30 V_{Th} [2] 8 V

31 If the resistance of R_L is doubled, what would the power delivered to R_L be? [1] 1.778 W

$$\textcircled{30} \quad P_{max} = \frac{V_{Th}^2}{4R_{Th}} = 4, \quad \frac{V_{Th}}{2R_{Th}} = 1\text{ A} \Rightarrow V_{Th} = 2R_{Th}$$

$$\frac{V_{Th}^2}{2V_{Th}} = 4 \Rightarrow V_{Th} = 8\text{ V}.$$

$$\textcircled{31} \quad \text{at } P_{max}, R_L = R_{Th}, \text{ if now } R_L = 2R_{Th}.$$

$$\text{then } P_L = V \cdot I = V_{Th} \cdot \frac{2R_{Th}}{R_{Th} + 2R_{Th}} \cdot \frac{V_{Th}}{3R_{Th}}$$

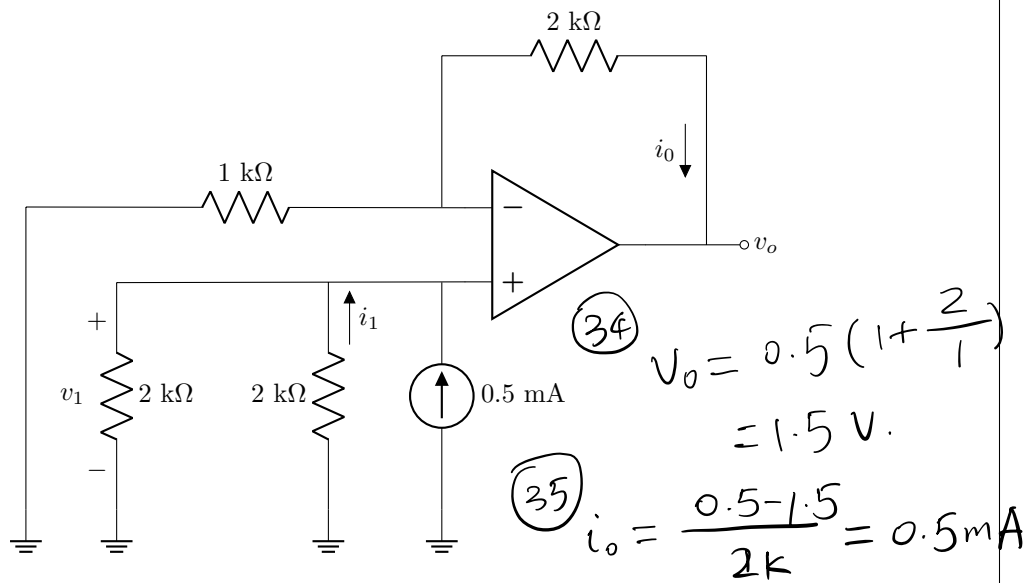
$$= \frac{V_{Th}^2}{9R_{Th}},$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = 4\text{ W} \Rightarrow \frac{V_{Th}^2}{R_{Th}} = 16$$

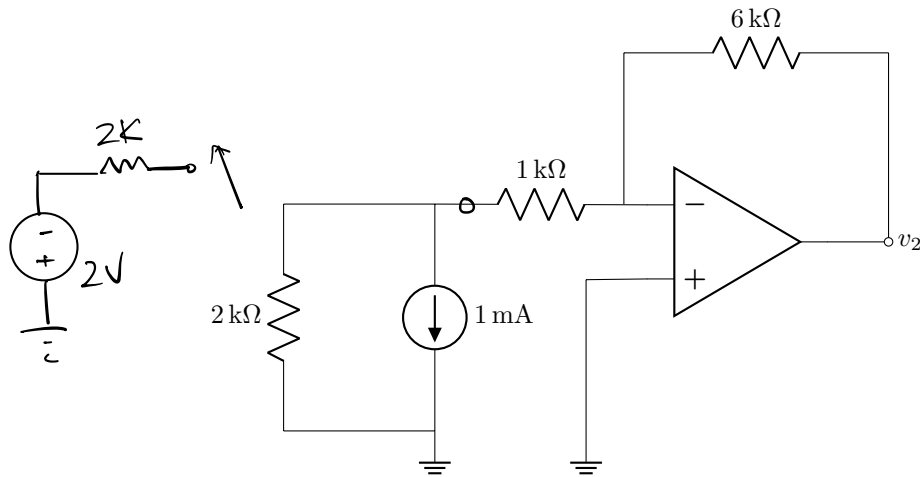
$$\Rightarrow P_L = \frac{16}{9} = 1.778\text{ W}$$

Questions 32 to 35 [4]

Assume that the op-amp is ideal.

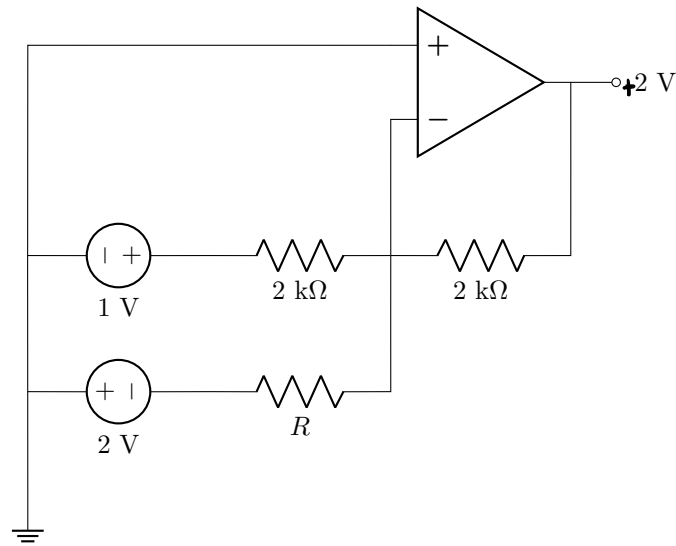


32	Determine i_1 . [1]	-0.25 mA
33	Determine v_1 . [1]	0.5 V
34	Determine v_o . [1]	1.5 V
35	Determine i_o . [1]	0.5 mA

Question 36 [2]Assume that the op-amp is ideal. Find v_2 .**36** v_2 [2]

4 V

$$v_2 = -(-2) \cdot \frac{6k}{3k} = 4 \text{ V.}$$

Question 37 [2]Assume that the op-amp is ideal. Determine R .**37** | R [2]

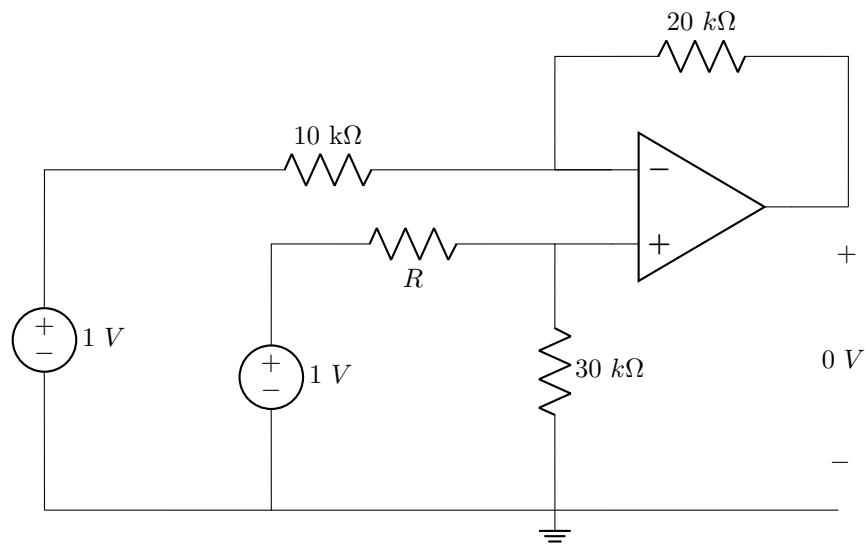
$$2 = - \left[\frac{2k}{2k} \cdot 1 + \frac{2k}{R} \cdot (-2) \right]$$

$$2 = -1 + \frac{2k}{R} \cdot 2$$

$$2R = -R + 4k$$

$$3R = 4k$$

$$R = 1.33 k\Omega$$

Question 38 [2]Determine the resistance R (assume that the op-amp is ideal).

38 R. [2]

15 kΩ

$$V_o = \frac{20k \left(1 + \frac{10k}{20k}\right)}{10k \left(1 + \frac{R}{30k}\right)} \cdot 1 - \frac{20k}{10k} \cdot 1 = 0$$

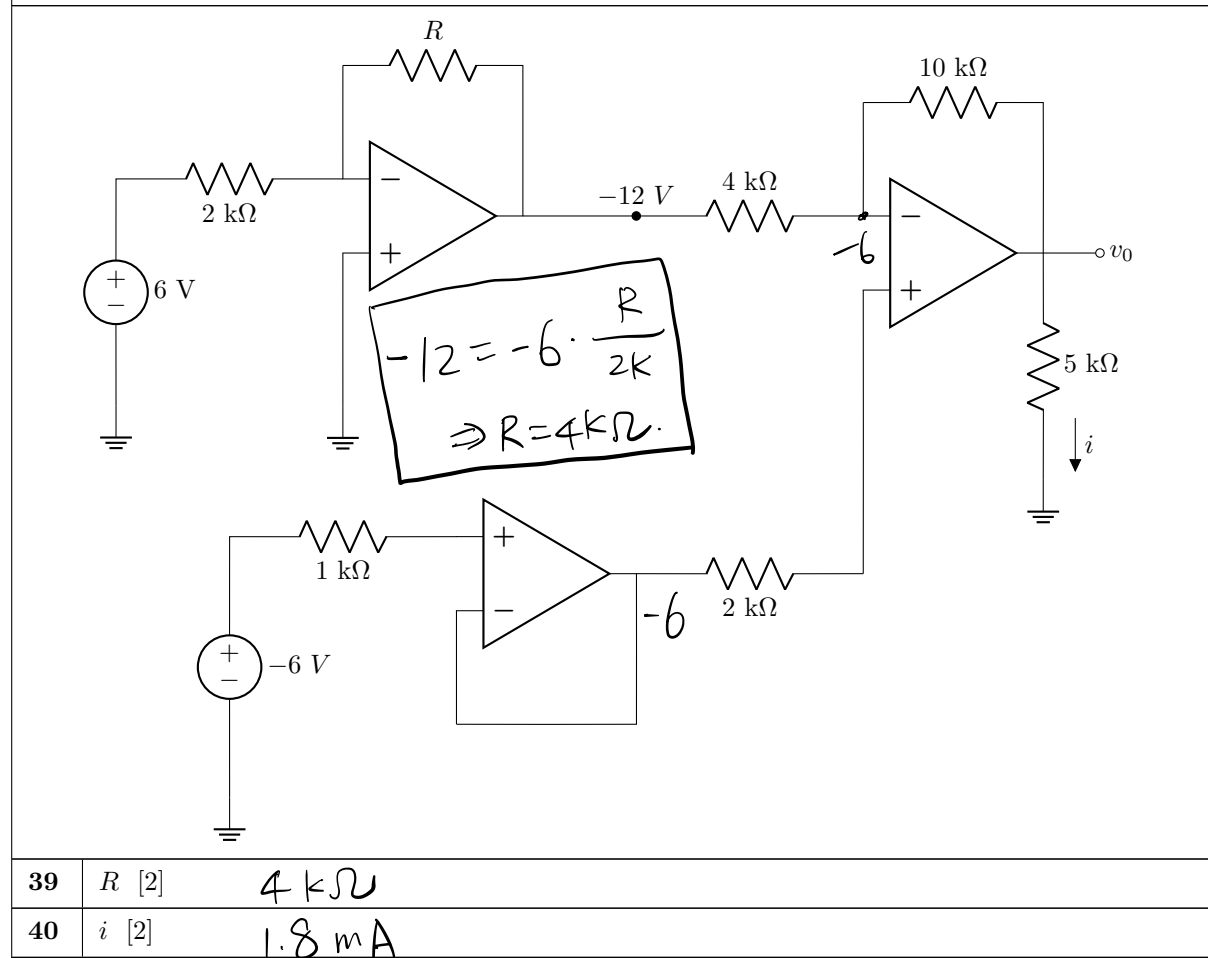
$$\frac{2 \times 1.5}{1 + \frac{R}{30k}} - 2 = 0$$

$$\frac{3}{1 + \frac{R}{30}} = 2 \Rightarrow 1 + \frac{R}{30} = 1.5$$

$$\Rightarrow R = 15 \text{ k}\Omega$$

Questions 39 to 40 [4]

Assume that the op-amps are ideal. Calculate the resistor R and the current i .



$$\frac{-12 - (-6)}{4k} = \frac{-6 - v_o}{10k}$$

$$\Rightarrow \frac{6}{4} = \frac{6 + v_o}{10}, \quad 60 = 24 + 4v_o$$

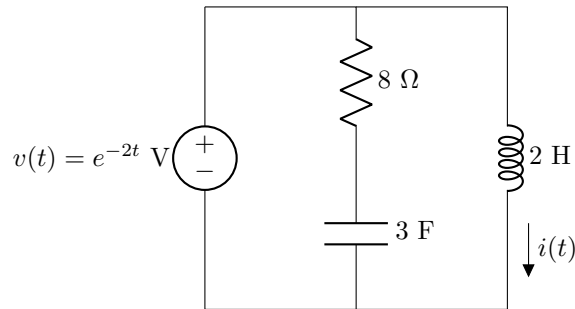
$$\Rightarrow v_o = 9V$$

$$i = \frac{9}{5k} = 1.8mA$$

Total Section 2: [31]

Section 3

Question 41 [2]



at

41 Calculate $i(t)$ between ~~$t = 0$ s~~ and $t = 3$ s when $i(0) = 2$ A. [2]

2.249 A

$$i(t) = 2 + \frac{1}{L} \int_0^t e^{-2\tau} d\tau$$

$$= 2 + \frac{1}{2} \cdot \left(-\frac{1}{2}\right) \int_0^t e^{-2\tau} d(-2\tau)$$

$$= 2 - \frac{1}{4} e^{-2\tau} \Big|_0^t$$

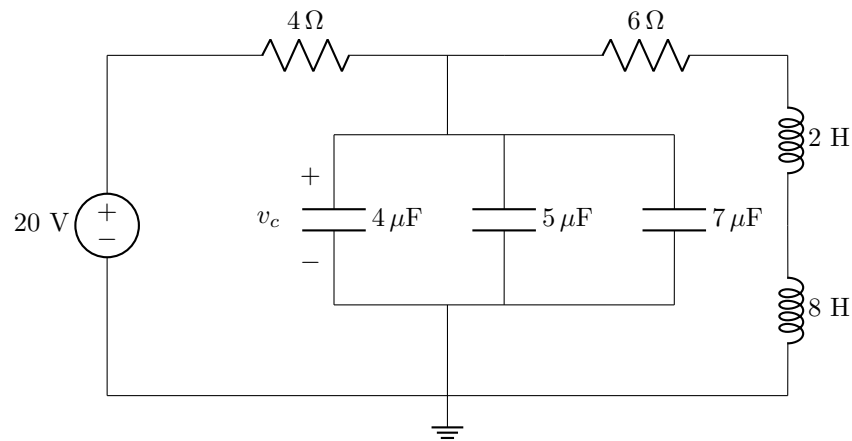
$$= 2 - \frac{1}{4} (e^{-2t} - 1)$$

$$= \frac{9}{4} - \frac{1}{4} e^{-2t}$$

$$i(3) = \frac{9}{4} - \frac{1}{4} e^{-6} = 2.249\text{ A}$$

Questions 42 to 43 [3]

The circuit below is under steady dc condition. Determine the voltage v_c across the $2\ \mu\text{F}$ capacitor and the total energy W_L stored in the inductors.

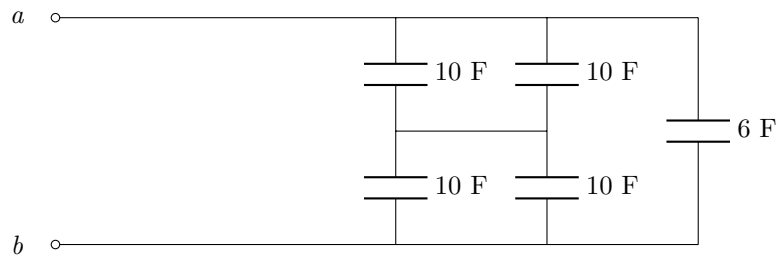


42	v_c [1]	12 V
43	W_L [2]	20 J

$$v_c = 20 \cdot \frac{6}{10} = 12 \text{ V.}$$

$$i_L = \frac{20}{10} = 2 \text{ A.}$$

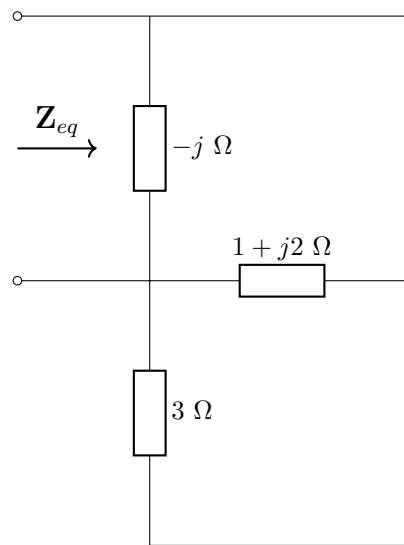
$$W_L = \frac{1}{2} L i_L^2 = \frac{1}{2} \times (8+2) \times 2^2 = 20 \text{ J}$$

Question 44 [2]

44 Calculate the equivalent capacitance C_{eq} with respect to terminals $a - b$. [2] 16F

$$\begin{aligned} C_{eq} &= [(10F || 10F) + (10F || 10F)] || 6F \\ &= (20F + 20F) || 6F \\ &= 16F \end{aligned}$$

Question 45 [2]



Three impedances
in parallel.

45 Find Z_{eq} . [2] $1.246 \angle -48.37^\circ \Omega$.

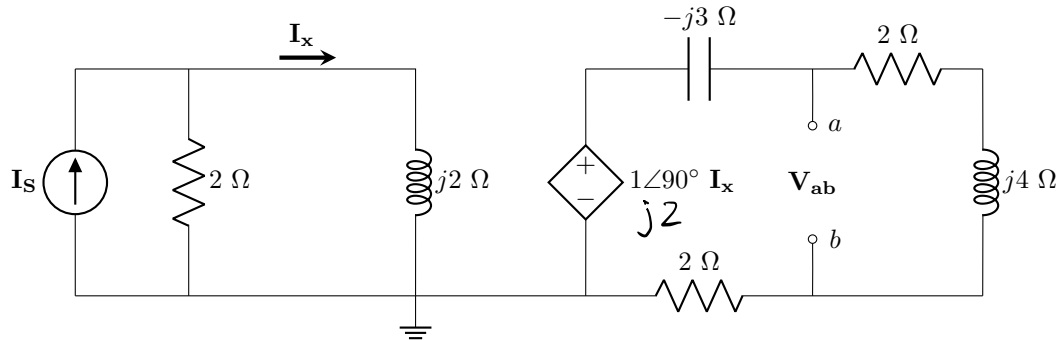
$$Z_1 = 3 \parallel -j = \frac{-j3}{3-j} = 0.3 - j0.9$$

$$Z_2 = 1 + j2$$

$$Z_{eq} = Z_1 \parallel Z_2 = \frac{(1+j2)(0.3-j0.9)}{1.3 + j1.1} = 1.246 \angle -48.37^\circ \Omega$$

Questions 46 to 47 [4]

If $\mathbf{I}_x = 2\angle 0^\circ$ A, find the phasor current $\mathbf{I}_S$ and phasor voltage $\mathbf{V}_{ab}$.



46 $\mathbf{I}_S$ [2] $2 + j2$ A

47 $\mathbf{V}_{ab}$ [2] $2.169 \angle 139.4^\circ$ V

$$\mathbf{I}_x = \mathbf{I}_S \cdot \frac{2}{2 + j2} = 2 \Rightarrow \mathbf{I}_S = 2 + j2 \text{ A}$$

$$\mathbf{V}_{ab} = j2 \cdot \frac{2 + j4}{4 + j} = 2.169 \angle 139.4^\circ \text{ V}$$

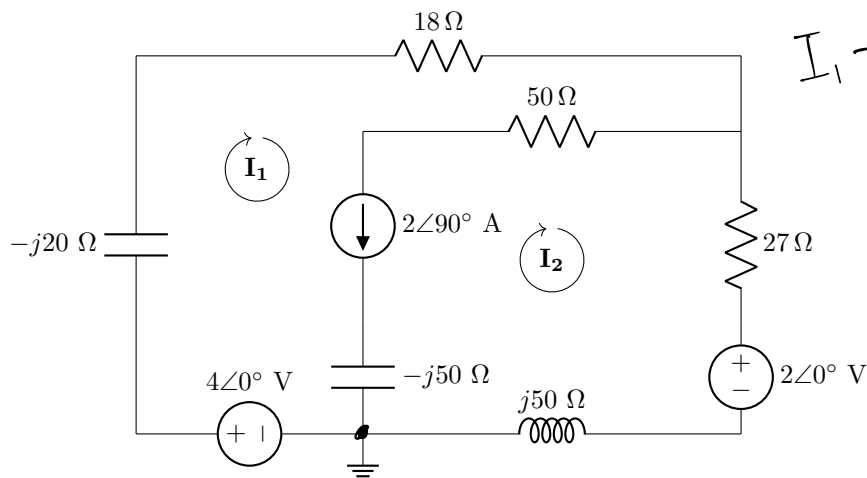
Questions 48 to 50 [4]

Use mesh analysis to set up two equations of the following form:

$$I_1 - I_2 = z_1,$$

$$aI_1 + bI_2 = 2\angle 0^\circ.$$

Find the unknown value z_1 and the unknown coefficients a and b .



48	z_1 [2]	$j2$
49	a [1]	$18 - j20$
50	b [1]	$27 + j50$

super mesh: $-4 - j20I_1 + 18I_1 + (27 + j50)I_2 + 2 = 0$

$$(18 - j20)I_1 + (27 + j50)I_2 = 2.$$

Questions 51 to 53 [4]	
Consider the following two sinusoidal waveforms:	
$v_1(t) = 20 \sin(\omega t + 60^\circ),$ $v_2(t) = -60 \cos(\omega t - 20^\circ).$	
51	Choose the correct statement. 1. v_1 leads v_2. 2. v_2 leads v_1. Write 1 if statement 1 is correct or 2 if statement 2 is correct. [1]
52	Determine the phase angle ϕ_d between the two sinusoids, with $\phi_d \in [0^\circ, 180^\circ]$. [2]
53	Use phasors to find $\mathbf{V} = \mathbf{V}_1 - \mathbf{V}_2$. [1]

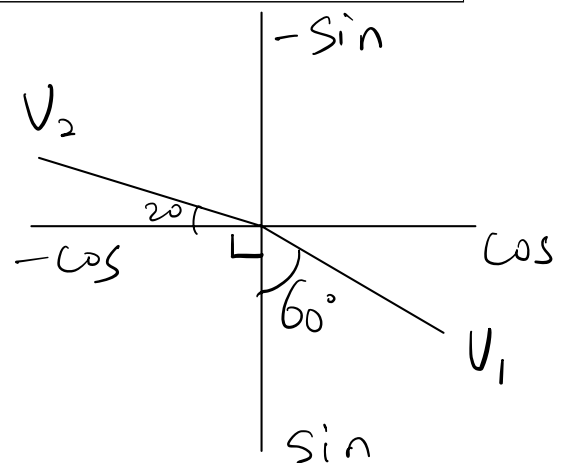
$$\phi_d = 20^\circ + 90^\circ + 60^\circ$$

$$= 170^\circ$$

$$\mathbf{V}_1 = 20 \angle -30^\circ$$

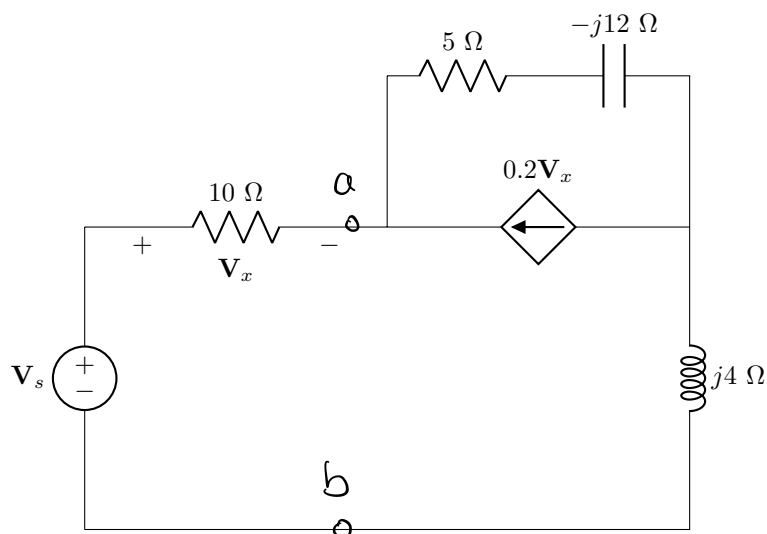
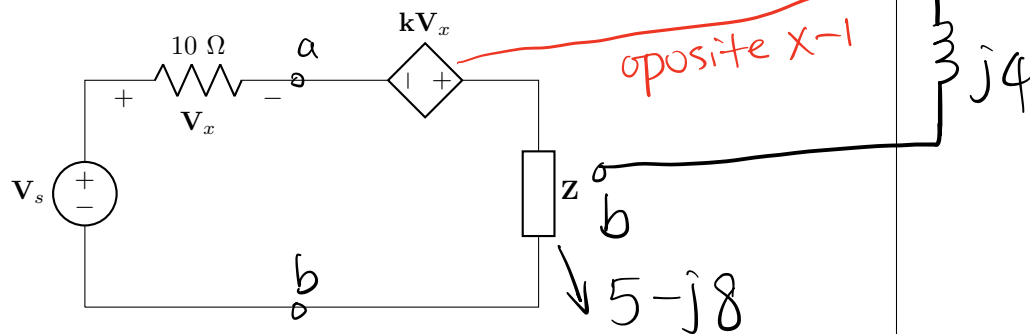
$$\mathbf{V}_2 = 60 \angle 160^\circ$$

$$\mathbf{V} = 40.45 \angle 164.9^\circ$$



Questions 54 to 55 [4]

Consider circuits **A** and **B** below. Transform circuit **A** such that it is equivalent to circuit **B**.

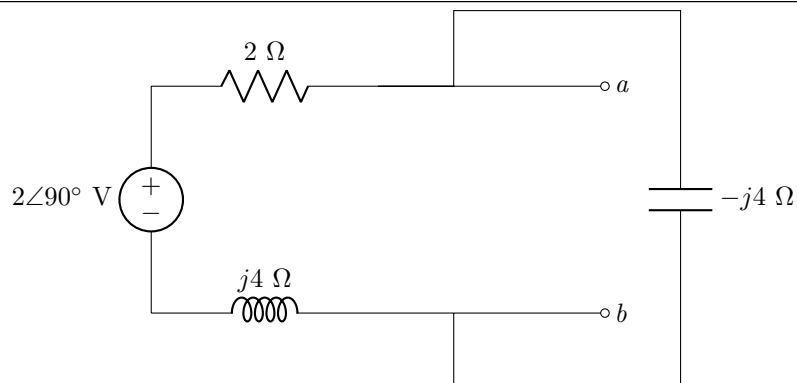
Circuit **A**Circuit **B**

54 Find **Z**. [2] $5 - j8\ \Omega$

55 Find **k**. [2] $-1 + j2.4$

Questions 56 to 57 [4]

Determine the Norton equivalent current $\mathbf{I}_N$ and the Norton equivalent impedance $\mathbf{Z}_N$ with respect to terminals a - b .



56	$\mathbf{I}_N$ [2]	$0.4 + j0.2 \text{ A}$
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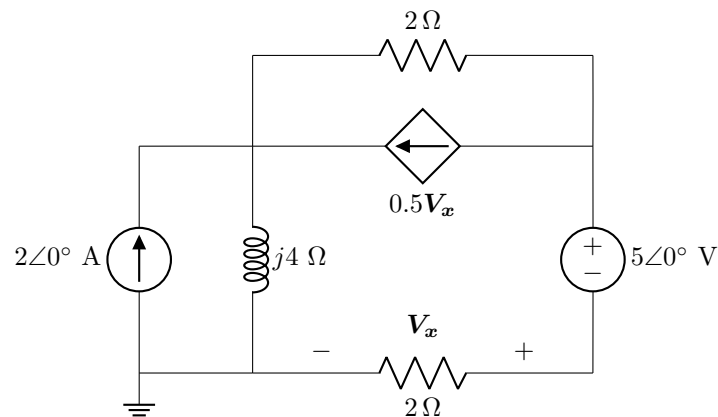
57	$\mathbf{Z}_N$ [2]	$8 - j4 \Omega$
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$$\mathbf{I}_N = \frac{j2}{2 + j4} = 0.4 + j0.2 \text{ A}$$

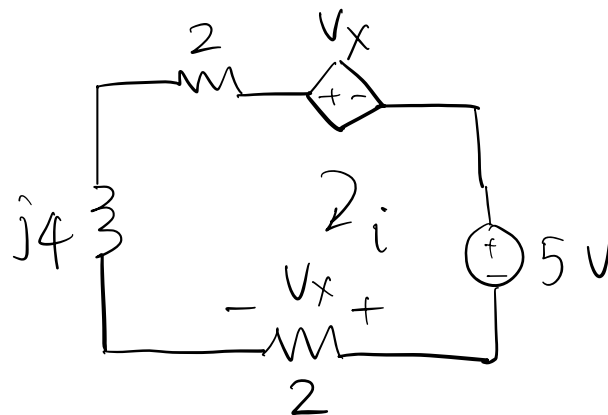
$$\mathbf{Z}_N = -j4 \parallel (2 + j4) = \frac{-j4(2 + j4)}{2} = 8 - j4 \Omega$$

Question 58 [2]

Use source transformation to find the exclusive contribution V_x^V from the independent voltage source to V_x .

58 V_x^V [2]

$$1.387 \angle 146.3^\circ \text{ V}$$



$$(4 + j4)i + V_x + 5 = 0, \quad V_x = 2i$$

$$(6 + j4)i = -5, \quad i = \frac{-5}{6 + j4}$$

$$V_x^V = 2i = \frac{-10}{6 + j4} = 1.387 \angle 146.3^\circ \text{ V}$$

Total Section 3: [31]**Grand Total : [93]**